

Detecting Weak Structure in Stochastic Particle Systems: A Matched-Counterfactual Multi-Collector Benchmark

Anonymous Authors

Abstract

We develop a matched-counterfactual instrument for testing whether mobile collectors exploit a weak latent transport field beyond capture induced by passive advection. Each signal episode is paired with a no-signal episode sharing its initial state, Brownian forcing, field nuisance variables, and policy randomness. Fixed-geometry contact is solved along exact piecewise specular paths, and exact ownership ties are event-keyed. A 12-seed exploratory calibration produced no positive simultaneous lower bound, while stationary and random controls descriptively matched or exceeded the local-flow policy. We therefore corrected the first-event estimand to subtract the stationary signal-null response and introduced a catchability ratio that separates drift strength from collector speed. A distinct full-state action-feasible oracle failed that preregistered first-event feasibility gate after one bounded state-input repair: first contacts occurred within a few steps, indicating endpoint saturation rather than a compute shortage. We then changed only the endpoint and stopping rule to count unique team captures through the inclusive end of step 67, one sensing-radius traversal at maximum collector speed. On eight fresh diagnostic seeds at $\alpha = 0.06$, the oracle exceeded stationary collectors by 9.375 particles on average (8/8 positive) and the true-field-only control by 7.5 (7/8 positive); all frozen correctness, direction, targeting, pairing, checksum, and artifact gates passed. This establishes diagnostic action-contingent headroom for the redesigned endpoint only. The shared-summary treatment remains unexecuted and its claim remains blocked. **No coordination effect, timestep convergence, power, confirmation, learning, or MARL result is claimed.**

1 Research question and novelty boundary

The intended paper asks one question: *with four collectors in the frozen unit-square task at $\alpha = 0.06$, does replacing each collector’s independent local velocity estimate with the same bounded team-mean estimate increase the number of unique particles captured through the inclusive end of step 67?* This comparison is presently blocked rather than answered: the redesigned endpoint has passed its upstream action-sensitivity gate, but coupled-noise timestep validation and a separate sharing preregistration must precede any shared-versus-independent execution.

Prior work already studies stationary capture in chaotic flow [1], a locally informed mobile collector capturing passive particles [2], local-communication collective sensing [4], learned and bandwidth-aware multi-agent communication [5, 6], and local-sensing reinforcement learning for active-particle foraging [3]. We therefore do not claim the first mobile particle collector, locally guided capture task, distributed flow-sensing framework, bounded communication method, particle-foraging problem, or local-sensing learning formulation. A fixed team mean is only a mechanism probe. The defensible target, if upstream gates eventually pass, is the causal effect of sharing that precisely defined statistic under matched sensing, action, and policy-capacity budgets, beyond passive advection and independent parallel search. Exact within-step intersection is likewise not a numerical novelty or a substitute for convergence of stochastic hitting statistics [7, 8].

2 Frozen system and local policy

The canonical domain is the unit square with reflecting boundaries. Four collectors act among $N = 256$ non-learning particles for $H = 400$ steps with $\Delta t = 0.02$ and $\sigma = 0.06$. Before capture, particle k follows

$$x_{k,t+1} = \mathcal{R}\left(x_{k,t} + \Delta t \alpha b_z(x_{k,t}) + \sigma \sqrt{\Delta t} \epsilon_{k,t}\right), \quad \epsilon_{k,t} \sim \mathcal{N}(0, I_2),$$

where the primary field is spatially uniform. Collector action $a_{i,t} \in \mathbb{R}^2$ is projected onto the unit ball before a common speed limit $v_{\max} = 0.12$ is applied:

$$y_{i,t+1} = \mathcal{R}\left(y_{i,t} + \Delta t v_{\max} \frac{a_{i,t}}{\max\{1, \|a_{i,t}\|_2\}}\right).$$

Collectors reset in ID order at $(.25, .25), (.25, .75), (.75, .25), (.75, .75)$. Particles are rejection-sampled from a dedicated initialization stream until strictly outside every initial capture disc. The episode pre-generates its full Brownian tensor and draws one field orientation from a separate stream.

The frozen `local_flow.v1` rule acts independently for each collector. It averages only causally valid visible-particle velocity slots, moves at unit normalized speed opposite that mean, and stops when none is valid. It has no density fallback and ignores the available teammate-position channel. Thus the nominal four-agent treatment is exactly four independent replicas, not a coordination treatment.

Each local observation contains the arena-normalized self position; up to $K = 32$ nearest free particles within radius 0.16; masks; and teammate positions relative to self in fixed ID order. A particle velocity is emitted only when that particle was visible to the same collector in both consecutive observations. The interface receives no field parameter, global state, future noise, or random generator.

3 Exact fixed-geometry contact and event-keyed ties

Endpoint contact is inadequate when either body reflects during a step. The implemented fixed-geometry rule retains each unfolded proposal, finds the union of particle and collector wall-hit fractions, and decomposes the step into piecewise-linear rectangular specular segments. On each interval, the earliest disc-point contact is the first admissible root of the relative-motion quadratic. Ownership is ordered by earliest contact fraction, then closest segment distance.

Any remaining exact tie is sampled from sorted stable collector IDs using `SeedSequence(s, t, k)`, where s is the scenario seed, t the one-based step, and k the stable particle ID. The decision therefore does not depend on previous random-number consumption, entity-array order, or unrelated events. The solver is exact for the piecewise-linear specular path inside one Euler step; it is not evidence that the discretized stochastic dynamics have converged as $\Delta t \rightarrow 0$. Growing aggregates retain deferred endpoint semantics and are excluded from the primary experiment.

4 First-event diagnostic history, physical axes, and provenance

Let $T_{s,\pi,c}^* \in \{1, \dots, H, H+1\}$ be the one-based first-interception step for scenario seed s , policy π , and condition c , with $H+1$ denoting no contact. The policy-specific signal-null gain is

$$G_s(\pi, \alpha) = \frac{T_{s,\pi,0}^* - T_{s,\pi,\alpha}^*}{H}.$$

Because the same signal can change passive flux, the corrected mechanism estimand subtracts the response of stationary collectors:

$$A_s(\pi, \alpha) = G_s(\pi, \alpha) - G_s(\text{stationary}, \alpha).$$

For the superseded first-event coordination comparison,

$$\Gamma_s(\alpha) = A_s(\text{shared}, \alpha) - A_s(\text{independent}, \alpha) = G_s(\text{shared}, \alpha) - G_s(\text{independent}, \alpha).$$

The stationary term cancels algebraically in Γ_s , but all four absolute signal/null outcomes and both passive-adjusted effects must still be reported so that passive flux and endpoint saturation remain visible.

The active redesigned endpoint instead lets $Y_s(\pi, \alpha)$ denote the number of unique particles captured by the team through the inclusive end of step 67. The episode continues after its first contact. The window is

$$L = \left\lceil \frac{0.16}{0.12 \times 0.02} \right\rceil = 67,$$

so a maximum-speed collector can traverse one sensing radius. The frozen upstream contrast is $d_s = Y_s(\text{oracle}, 0.06) - Y_s(\text{stationary}, 0.06)$, with practical threshold $E[d_s] \geq 4$, corresponding to one additional capture per collector. A mandatory targeting contrast replaces stationary with the true-field-only upstream control. Any later coordination estimand will be the matched seed-level difference $Y_s(\text{shared}, 0.06) - Y_s(\text{independent}, 0.06)$; its minimum relevant effect, inference procedure, and seed budget have not been frozen.

Two dimensionless coordinates are required:

$$\rho = \frac{\alpha \sqrt{\Delta t}}{\sigma}, \quad \kappa = \frac{\alpha}{v_{\max}}.$$

Here ρ compares drift with per-step Brownian displacement, whereas κ compares particle drift with collector control authority. Under the canonical $\Delta t = 0.02$, $\sigma = 0.06$, and $v_{\max} = 0.12$, the old $\rho = 2$ point has $\kappa \approx 7.07$; it is therefore a poor catchability diagnostic. The bounded oracle gate instead used $\kappa \in \{0.25, 0.5, 1.0\}$. Complete scenario-seed blocks remain the sampling units. Any future boundary requires a separately preregistered grid, minimum effect, simultaneous lower bound, and independent confirmation seeds.

The matched runner rejects a pair unless its environment configurations differ only in signal strength and the null strength is zero. It verifies identical particle and collector initialization, Brownian tensors, and field nuisance variables. It records SHA-256 digests of those inputs, policy randomness, and the event-key specification. At zero signal, full records must be identical after ignoring only the condition label.

Each pair is schema-validated before an immutable write. The runner produces null and signal JSONL trajectories, a pair summary, and a manifest containing the repository commit and dirty flag, environment-config digest, runtime command and versions, policy privilege flag, and artifact byte counts and SHA-256 digests. The bounded pilots retain compact pair summaries and aggregate analyses; they are not a released trajectory dataset.

5 Studentized simultaneous bootstrap

Rows of effect matrix G are scenario-seed blocks and columns are the five nonzero preregistered values $\rho \in \{0.1, 0.25, 0.5, 1, 2\}$. Each bootstrap draw resamples complete rows, retaining dependence across the grid. With observed means $\bar{G}_j$ and standard errors $\hat{\text{se}}_j$, the common correction is

$$q_{.95} = Q_{.95} \left[\max_j \frac{\bar{G}_j - \bar{G}_j^*}{\hat{\text{se}}_j^*} \right], \quad L_j = \bar{G}_j - q_{.95} \hat{\text{se}}_j.$$

The frozen implementation uses 10,000 draws and bootstrap seed 73,031.

Synthetic null calibration used 2,000 trials, 2,000 bootstrap draws per trial, 12 rows, and five columns. The familywise false-crossing rate was 0.040 under a correlated continuous null (Wilson 95% upper bound 0.0495) and 0.006 under a zero-inflated symmetric discrete null (upper bound 0.0105). Both met the predeclared acceptance rule that this upper bound not exceed 0.07. This checks only synthetic calibration, not the scientific hypothesis.

6 Twelve-seed exploratory calibration

SPS-P02 used scenario seeds 1001–1012 to reproduce the frozen calibration after the exact contact repair. These seeds were assigned to runtime, variance, censoring, oracle-feasibility, and contact diagnostics. They *cannot* update SPS-C01. Table 1 documents the exploratory calculation.

Table 1: Exploratory `local_flow_v1` calibration. Each row uses the same 12 pilot seeds; LCBs are simultaneous one-sided 95% bounds.

ρ	Mean D	Paired SE	Simultaneous LCB
0.10	0.00563	0.00339	-0.01533
0.25	0.00375	0.00457	-0.02452
0.50	0.00708	0.00423	-0.01908
1.00	0.00896	0.00459	-0.01944
2.00	0.01104	0.00514	-0.02079

The studentized maximum critical value was 6.187. Every LCB was negative, so the exploratory rule returned right-censoring beyond the largest tested point, $\rho = 2$. Because the seeds are calibration-only, this neither supports nor refutes a finite boundary and is not a boundary estimate.

Table 2 shows the strongest-signal controls. Stationary, pre-generated random, and privileged upstream policies descriptively matched or exceeded the four-independent local-flow mean. The one-collector row is not capacity-matched and is diagnostic only. No inferential comparison among these controls was authorized.

Table 2: Exploratory controls at $\rho = 2$, each with 12 calibration seeds.

Policy / treatment	Mean D	Median D
Stationary, $M = 4$	0.01313	0.00625
Pre-generated random, $M = 4$	0.01292	0.00875
Privileged upstream oracle, $M = 4$	0.01292	0.00625
<code>local_flow_v1</code> , $M = 4$ independent	0.01104	0.00375
Coverage, $M = 4$	0.00771	0.00625
Density greedy, $M = 4$	0.00625	0.00250
<code>local_flow_v1</code> , $M = 1$, diagnostic	0.06938	0.01625

Contact-repair comparison. SPS-P01 guarded every reflected path pair; SPS-P02 used exact specular segments. Across the same 144 matched pairs, guarded diagnostics fell from 28,929 to zero and zero of 144 first-interception outcomes changed. This supports the regression repair, not independent scientific replication.

Power status. The earlier provisional 24-seed plan targeted a half-width of 0.05, which is coarser than the exploratory effects of approximately 0.004–0.011. It is not a scientifically usable confirmation budget and SPS-E01 remains blocked. No simulation-based power study has been run for the passive-adjusted or coordination estimand. **PENDING: After a redesigned endpoint passes its oracle and timestep gates, freeze a minimum relevant effect and calibrate type-I error and power over complete scenario-seed blocks before authorizing confirmation.**

7 First-event oracle failure on the catchable grid

The legacy `privileged_upstream_oracle` is renamed conceptually as a *true-field upstream control*: it knows the true field direction and moves upstream, but it neither observes every target position nor computes a feasible intercept. It is not an interception oracle. The distinct full-state oracle observes all free-particle positions and the known current deterministic drift, assigns distinct feasible targets to collectors over a bounded receding horizon, obeys the same action and speed limits, and receives no future Brownian increments.

SPS-P03 exposed an implementation error: the first oracle version treated the last realized Brownian displacement as predictive motion state. The only permitted state-input repair replaced that quantity with current deterministic drift and was preregistered as SPS-P04. The corrected oracle still failed every nonzero

feasibility point (Table 3). This is an exploratory diagnostic gate, not a significance test or confirmatory claim.

Table 3: SPS-P04 corrected full-state-oracle gate on eight diagnostic seeds. Passing required positive mean passive-adjusted gain, at least six positive seed directions, and favorable absolute comparisons. No point passed.

α	κ	Mean A_s	Positive seeds	Gate
0.03	0.25	-0.00844	0/8	Fail
0.06	0.50	-0.00563	1/8	Fail
0.12	1.00	-0.01063	2/8	Fail

The failure is consistent with first-event saturation in the canonical reset. Across the diagnostic grid, stationary collectors averaged 5.78 steps to first contact and the corrected oracle averaged 3.00; every episode was uncensored, so the 400-step horizon never bound. The early contacts showed no wall involvement. The oracle can improve absolute null performance while still having a negative policy-minus-stationary signal response because passive signal transport benefits stationary collectors strongly and leaves little action-contingent headroom in the signal arm. Thus the current first-event endpoint and initial geometry do not identify the proposed mechanism.

Per the preregistered stop rule, SPS-P04 blocks SPS-C03 and all downstream scientific execution. A later run that exceeded the one-repair allowance is preserved for audit but procedurally excluded from every claim, manuscript decision, design choice, and power calculation; none of its outcomes are used here.

8 Fixed-horizon action-sensitivity gate

SPS-WO-05 changed only the endpoint and episode stopping rule. The unit-square arena, canonical collector lattice, 256 particles, four collectors, field, diffusion, capture geometry, actions, policies, and matched random streams were unchanged. Captured particles were counted once by stable particle ID, episodes continued after first contact, step 67 was included, and step 68 was excluded. The frozen diagnostic used fresh scenario seeds 2001–2008, field strengths $\alpha \in \{0, 0.06\}$, and stationary, true-field-only, and full-state-oracle policies. These seeds and outcomes are permanently ineligible for confirmation.

At $\alpha = 0.06$, the oracle-minus-stationary seed contrasts were

$$(11, 13, 9, 12, 7, 7, 11, 5),$$

with mean 9.375, median 10, sample standard deviation 2.825, and positive direction in 8/8 seeds. The descriptive paired-bootstrap 95% interval was [7.5, 11.25]. Oracle-minus-true-field-control contrasts were

$$(6, 12, 6, 15, 8, 3, 11, -1),$$

with mean 7.5, median 7, sample standard deviation 5.155, and positive direction in 7/8 seeds; the descriptive paired-bootstrap 95% interval was [4.125, 10.875]. All frozen gate components passed: deterministic correctness and execution, mean oracle gain of at least four, positive primary direction in at least six seeds, positive mean targeting contrast, matched streams, checksums, and artifact completeness. The intervals are descriptive, not confirmatory.

The valid package contains all 48 policy-condition episodes and 3,216 environment steps. A process-control race caused an earlier identical process to finish after it had been prematurely classified as invalid. Its episode and event files are bitwise identical to the canonical R1 package; it is retained only for audit and is neither pooled, counted as replication, nor used to improve precision. R1 is the sole evidence package.

This result repairs the upstream feasibility diagnosis: unlike time to first contact, fixed-horizon unique yield leaves measurable room for legal actions to matter. It does not show that local observations are sufficient, that sharing helps, or that any learned multi-agent policy can exploit this headroom.

9 AAMAS relevance failure and blockers

The current treatment is not yet a multi-agent contribution. Teammate positions are observed but unused; the four collectors apply independent copies of one scripted rule; and no learned policy is evaluated. The pilot therefore concerns replicated local control and passive transport, not cooperation, communication, strategic interaction, or MARL. No coordination or MARL claim is supported.

The initial scientific no-go remains relevant to the superseded endpoint. At $\rho = 2$, passive stationary and uninformed random controls descriptively matched or exceeded `local_flow_v1`; at the catchable points, the repaired full-state oracle failed its passive-adjusted first-event gate. The fixed-horizon diagnostic now establishes action-contingent headroom, but not a multi-agent effect. SPS-C03 therefore remains blocked and the shared-summary policy, although implemented and unit-tested, has not been run as a scientific treatment. Coupled Brownian aggregation utilities are tested, but no timestep-convergence experiment has been executed. No empirical power calculation, independent confirmation, learned policy, IPPO/MAPPO baseline, or MARL experiment exists. Growing-aggregate physics and aggregation analysis remain later work.

The implementation currently has 89 passing tests covering deterministic reset, causal observations, zero-signal identity, exact specular contacts, event-order and event-key invariance, schema validation, immutable writes, paired provenance, frozen policies, row-order-invariant inference, bounded team-summary invariance, full-state-oracle microcases, and exact Brownian aggregation, as well as unique-yield counting, inclusive step-67 evaluation, post-contact continuation, and matched yield-run provenance. These are engineering checks only. SPS-C01 remains proposed but inactive, SPS-C03 is blocked, and **no detectability boundary is supported by SPS-P01 or SPS-P02**.

10 Limitations

Any eventual boundary is policy-, horizon-, field-, geometry-, grid-, and resource-relative rather than a universal physical critical point. Common random numbers reduce variance only under causally coherent pairing. Exact within-step specular contact does not replace a timestep-convergence study. The canonical first-contact endpoint is saturated by initial geometry and passive transport. The fixed-horizon endpoint clears an eight-seed diagnostic oracle gate but has not passed stochastic timestep validation. AAMAS fit requires an isolated multi-agent mechanism, not merely four collectors or a fixed statistic whose empirical effect has not been tested. **PENDING: Run the separately preregistered coupled-noise timestep diagnostic for unique yield; only if it passes, freeze and run a bounded shared-versus-independent diagnostic before any power, learning, or confirmation experiment.**

References

- [1] Wang et al. Particle capture in a model chaotic flow. *Physical Review E*, 2021. <https://doi.org/10.1103/PhysRevE.104.064203>.
- [2] Wang et al. Mobile-collector capture of particles in a chaotic flow. *PLOS ONE*, 2025. <https://doi.org/10.1371/journal.pone.0329766>.
- [3] R. C. Löffler, E. Panizon, and C. Bechinger. Collective foraging of active particles trained by reinforcement learning. *Scientific Reports*, 2023. <https://doi.org/10.1038/s41598-023-44268-3>.
- [4] H. Hornischer et al. CIMAX: Collective information maximization in robotic swarms using local communication. arXiv:1903.05444, 2019. <https://arxiv.org/abs/1903.05444>.
- [5] J. Foerster et al. Learning to communicate with deep multi-agent reinforcement learning. *NeurIPS*, 2016. https://proceedings.neurips.cc/paper_files/paper/2016/hash/c7635bfd99248a2cdef8249ef7bfbef4-Abstract.html.
- [6] T. Wang et al. Learning efficient multi-agent communication: An information bottleneck approach. *ICML*, 2020. <https://proceedings.mlr.press/v119/wang20i.html>.

- [7] E. Gobet. Weak approximation of killed diffusion using Euler schemes. *Stochastic Processes and their Applications*, 2000. [https://doi.org/10.1016/S0304-4149\(99\)00109-X](https://doi.org/10.1016/S0304-4149(99)00109-X).
- [8] M. B. Giles. Multilevel Monte Carlo path simulation. *Operations Research*, 2008. <https://doi.org/10.1287/opre.1070.0496>.